Question

μ-oxo-bis[bis(diethyldithiocarbamato)oxomolybdenum(V)] (Complex **A**) is a molybdenum complex containing sulfur ligands. Studying the properties of such substances helps us understand the structure and performance of various molybdenum enzymes in biological systems. Experimental results show that the solution of **A** does not obey the Lambert-Beer law, indicating the existence of the following disproportionation equilibrium:

$$[\mathrm{MoO(S_2CNEt_2)_2}]_2\mathrm{O}\ (\mathbf{A}) \rightleftharpoons \mathrm{MoO(S_2CNEt_2)_2}\,(\mathbf{B}) + \mathrm{MoO_2(S_2CNEt_2)_2}\,(\mathbf{C})$$

Solutions of **A** were prepared with initial concentrations of $c_1 = 2.07 \times 10^{-4}\mathrm{mol{\cdot}L^{-1}}$ and $c_2 = 1.55 \times 10^{-4}\mathrm{mol{\cdot}L^{-1}}$, and allowed to reach disproportionation equilibrium. The absorbances were measured at $\lambda = 510\mathrm{nm}$ (the absorption band of the bridging oxygen) in a $b = 1.00\ \mathrm{cm}$ cuvette, yielding $A_1 = 0.616$ and $A_2 = 0.385$, respectively. It is known that at $\lambda = 510\mathrm{nm}$, the molar absorptivity of **B** is ε_B=715$\mathrm{L{\cdot}mol^{-1}{\cdot}cm^{-1}}$, and its contribution to absorbance cannot be ignored. In contrast, the molar absorptivity of **C** is very small, and its effect on absorbance can be neglected. Please calculate the equilibrium constant K of the disproportionation reaction and the true molar absorptivity ε_A of **A**.

A. $K = 2.5 \times 10^{-3}\ \mathrm{mol \cdot L^{-1}}$

$\varepsilon_\mathrm{A} = 2.0 \times 10^4\ \mathrm{L{\cdot}mol^{-1}{\cdot}cm^{-1}}$

B. $K = 4.0 \times 10^{-3}\ \mathrm{mol \cdot L^{-1}}$

$\varepsilon_\mathrm{A} = 6.1 \times 10^3\ \mathrm{L{\cdot}mol^{-1}{\cdot}cm^{-1}}$

C. $K = 1.8 \times 10^{-3}\ \mathrm{mol \cdot L^{-1}}$

$\varepsilon_\mathrm{A} = 2.5 \times 10^4\ \mathrm{L{\cdot}mol^{-1}{\cdot}cm^{-1}}$

D. $K = 4.6 \times 10^{-4}\ \mathrm{mol \cdot L^{-1}}$

$\varepsilon_\mathrm{A} = 1.2 \times 10^4\ \mathrm{L{\cdot}mol^{-1}{\cdot}cm^{-1}}$

E. $K = 1.8 \times 10^{-4}\ \mathrm{mol \cdot L^{-1}}$

$\varepsilon_{\mathrm{A}} = 2.5 \times 10^4\ \mathrm{L{\cdot}mol^{-1}{\cdot}cm^{-1}}$

F. $K = 4.0 \times 10^{-4}\ \mathrm{mol \cdot L^{-1}}$

$\varepsilon_{\mathrm{A}} = 6.1 \times 10^3\ \mathrm{L{\cdot}mol^{-1}{\cdot}cm^{-1}}$

G. $K = 2.5 \times 10^{-4}\ \mathrm{mol \cdot L^{-1}}$

$\varepsilon_{\mathrm{A}} = 2.0 \times 10^4\ \mathrm{L{\cdot}mol^{-1}{\cdot}cm^{-1}}$

H. $K = 4.6 \times 10^{-3}\ \mathrm{mol \cdot L^{-1}}$

$\varepsilon_{\mathrm{A}} = 1.2 \times 10^4\ \mathrm{L{\cdot}mol^{-1}{\cdot}cm^{-1}}$

Answer

Correct Answer: C

Detailed Explanation

Let the equilibrium concentrations of **B** in the two sets of experiments be x_1 and x_2, respectively. Then, the following equations are established:

$$K = x_1^2/(c_1 - x_1)$$

$$K = x_2^2/(c_2 - x_2)$$

$$A_1 = \varepsilon_{\mathrm{A}} b(c_1 - x_1) + \varepsilon_{\mathrm{B}} b x_1$$

$$A_2 = \varepsilon_{\mathrm{A}} b(c_2 - x_2) + \varepsilon_{\mathrm{B}} b x_2$$

CHECKPOINT 1 PTS

$$K = x_1^2/(c_1 - x_1)$$

CHECKPOINT 1 PTS

$$K = x_2^2/(c_2 - x_2)$$

CHECKPOINT 1 PTS

$$A_1 = \varepsilon_{\mathrm{A}} b(c_1 - x_1) + \varepsilon_{\mathrm{B}} b x_1$$

CHECKPOINT 1 PTS

$$A_2 = \varepsilon_{\mathrm{A}} b(c_2 - x_2) + \varepsilon_{\mathrm{B}} b x_2$$

The solution yields:

$$K = 1.8 \times 10^{-3}\ \mathrm{mol} \cdot \mathrm{L}^{-1}$$

CHECKPOINT 2 PTS

$$K = 1.8 \times 10^{-3}\ \mathrm{mol} \cdot \mathrm{L}^{-1}$$

$$\varepsilon_{\mathrm{A}} = 2.5 \times 10^{4}\ \mathrm{L{\cdot}mol^{-1}{\cdot}cm^{-1}}$$

CHECKPOINT 2 PTS

$$\varepsilon_{\mathrm{A}} = 2.5 \times 10^{4}\ \mathrm{L{\cdot}mol^{-1}{\cdot}cm^{-1}}$$